Inbrev

Hack Apertus 2026 · Track 2B · Team Apertool

The practice inbox that never leaves the practice.

Inbrev sorts the inbox of a Swiss GP practice by handling path and deadline, with Apertus running on a box in the practice. Every field it extracts quotes its source.

▶ Watch the 2-min demo

Code on GitHub [TBD link]

solo builder · fictional data only

THE PROBLEM

Before the first patient, there is the inbox.

Lab reports, refill requests, insurer forms, referrals and faxes in German, French and Italian all land in one queue. A medical practice assistant (MPA) reads them one by one.

9.5 h / week

of administration for Swiss GPs and paediatricians in 2025, up from 8.6 h in 2020.

Zeller, Fischer, Hernandez, *Praxis* 2026 · 5th Workforce Study, n = 1,776

54 min / day

that practice-based doctors spend on requests from authorities and insurers alone.

gfs.bern for FMH, 2025 · N = 1,532 doctors

Art. 321

Swiss Criminal Code: professional secrecy binds doctors **and their assistants**. Every message in that inbox is covered.

SCC, SR 311.0 · health data are sensitive personal data under FADP Art. 5

HOW IT WORKS · PURPOSEFUL USE OF AI

Sorted by who handles it and by when. Every field quotes its source.

01 · READ

Every channel

E-mail, HIN mail, web forms and scanned faxes (OCR). German, French, Italian and English.

02 · ROUTE

9 handling paths

Report, appointment, refill, insurer, referral, certificate, billing, marketing, other. Plus an administrative deadline and who takes it: MPA, doctor or admin.

03 · PROVE

Quote or drop

Every field carries the exact words it came from. Code checks that the quote is in the original. If Inbrev is unsure, **a human decides**.

Safety net A message that may describe acute symptoms skips all automation and goes straight to a person. Inbrev does not judge it.

Apertus reads free text in four languages. Everything code can check (AHV numbers, dates, insurer names, quotes) is checked by code. Small local models do best on narrowly cut tasks (Xu et al., *JMIR* 2026, CHUV).

DEMO

One click. The possible emergency on top, the rest by path and deadline.

Inbox · sorted — 8 of 30 shown

HUMAN NOW	**Heidi Zimmermann** „Dringend“ – Schmerzen in der Brust	person
TODAY	**Labor Rheintal AG** Report received · Lukas Weber	doctor
THIS WEEK	**Fax · KV Beispiel AG (fict.)** Insurer · Kostengutsprache Physio	admin
THIS WEEK	**Marc Rochat · FR** Refill · Renouvellement ordonnance	MPA
THIS WEEK	**Tobias Frei · web form** Certificate · Arbeitsunfähigkeit	MPA
NO DEADLINE	**Dr. P. Bernasconi · IT** Referral · Invio paziente	MPA
YOU DECIDE	**Peter Huber** Two topics in one message – not sure	MPA

Fax · OCR · German · route **insurer** → admin · deadline **08.10.2026**

```
KOSTENGUTSPRACHE – RÜCKFRAGE
Versicherte Person: Anna Meier, 22.08.1979,
AHV 756.1234.9876.14
Wir benötigen bis 08.10.2026 die ärztliche
Verordnung für 9 Sitzungen Physiotherapie. Referenz
KG-2026-55813.
```

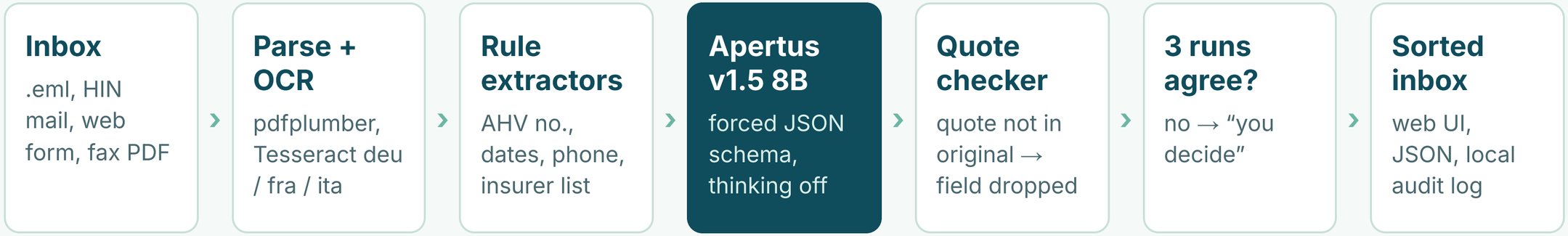
Patient	**Anna Meier** “Anna Meier”
Date of birth	**1979-08-22** “22.08.1979”
AHV no.	**756.1234.9876.14** “format check: code”
Deadline	**2026-10-08** “bis 08.10.2026”
Insurer ref.	**KG-2026-55813** “Referenz KG-2026-55813”

Mock-up with messages from the fictional seed set; no real patient data anywhere. [TBD: replace with screenshot of the real app]

ARCHITECTURE · SOVEREIGN DEPLOYABILITY

Pull the cable. Inbrev keeps sorting.

INSIDE THE PRACTICE · NO OUTBOUND CONNECTION AT RUN TIME



On-prem box	Mini PC with GPU in a group practice
Air-gapped	Model host offline; a mail gateway hands messages in
Swiss cloud	Option for single practices, Swiss hosting
CPU only	GGUF Q4 for judges without a GPU

```
$ docker run --rm --network none \
    -v ./inbox:/inbox inbrev triage /inbox
[TBD: paste real output of the offline run]

# downloaded at build time only:
# weights (SHA-256), images, OCR data
```

EVALUATION · TECHNICAL RIGOUR

Recall first: no acute message may slip through.

Target: recall ≥ 0.95 for “safety” and “today”, with the share of messages handed to a person disclosed.

Setup	Macro-F1 9 paths	Recall safety	Recall “today”	Runs automatically
Keyword rules	TBD	TBD	TBD	100 %
TF-IDF + log. regression	TBD	TBD	TBD	100 %
Other open 7–8B model	TBD	TBD	TBD	100 %
Apertus 8B, plain prompt	TBD	TBD	TBD	100 %
Apertus 70B (CSCS)	TBD	TBD	TBD	100 %
Inbrev (8B + checks + abstain)	TBD	TBD	TBD	TBD

Also reported: field F1, share of fields with a valid quote, and every metric split by language (DE / FR / IT) and writing level (clean / typos / non-native) and by channel (text / fax OCR).

Gold set: ~300 fictional messages

- Labels are fixed before any text is written.
- Text by two other model families, by hand, and as scanned faxes. Apertus never writes or labels test data.
- Fictional people; AHV numbers with an invalid check digit.
- ≥ 60 messages labelled twice, 48 h apart, blind: κ TBD
- Open data, CDLA-Permissive-2.0.

VALUE, COST & SCALABILITY

Swiss francs per 1,000 messages, on hardware the practice owns.

ON-PREM GPU

Apertus 8B · RTX 5070 Ti

CHF TBD

per 1,000 messages (power + depreciation)

Latency: TBD s / message

Energy: TBD Wh / message

CPU ONLY

GGUF Q4, no GPU

CHF TBD

per 1,000 messages

Latency: TBD s / message

Fine for an overnight batch? TBD

REFERENCE

Swiss cloud API

≈ CHF 3.35

per 1,000 requests of 3,000 input + 500 output tokens

Inbrev tokens / message: TBD

Up to 3 runs per message

From CHF 0.00335 per request, blog estimate (Pupel, devs-group, 2026).

ASSUMPTION

What it buys: 60 messages a day × 1 minute each × CHF 50/h MPA time ≈ CHF 50 a day. To be checked in interviews with practices.

INTENDED PURPOSE · IMPLEMENTATION FEASIBILITY

Administrative by design. Inbrev routes, it never triages.

Inbrev does

- ✓ Sort by handling path and administrative deadline
- ✓ Extract patient and case fields, each with its quote
- ✓ Hand anything that may be acute to a person, unranked
- ✓ Say “you decide” when it is unsure

Inbrev never does

- ✗ Rank messages by clinical severity
- ✗ Assess symptoms or suggest care
- ✗ Make a decision a person has not seen
- ✗ Send patient data out of the practice

Why the line sits here: MDCG 2019-11 Rev.1 (2025) treats software that automatically reprioritises by clinical criticality based on patient data as medical device software; software that only stores, communicates or searches is not. Switzerland classifies via MepV Art. 15 → EU MDR Annex VIII. Privacy: FADP Art. 5, 7 and 22, and SCC Art. 321.

BUSINESS MODEL & ROADMAP

A module for the software practices already use.

Who uses it	The MPA, first thing in the morning
Who pays	The practice owner or the doctors' network
Price	CHF 150–250 per practice and month HYPOTHESIS
Channel	As a module inside HIN mail or practice software, not one more window
Open core	Hackathon code Apache-2.0. Revenue from operation, connectors, support and updates
Later	Germany and Austria: same logic, same secrecy duty (§ 203 StGB)

OCT 2026 · HACK APERTUS
Open gold set, full evaluation, offline proof, CHF per 1,000 messages

NOV – DEC 2026
5–10 interviews with MPAs and practice owners: real volumes, categories, fax share

Q1 2027
Pilot in one group practice on its own box; mail and practice-software connectors

Q2 2027
Written intended purpose, impact assessment, classification review before first sale

Inbrev

Team Apertool · Niklas Wichter, solo builder

The practice inbox that never leaves the practice.

We are looking for one Swiss group practice to pilot Inbrev on its own hardware in Q1 2027.

Contact: `[TBD e-mail]`

Demo ▶ 2-min video

Code `[TBD GitHub link]`

Data `[TBD Hugging Face link]`